

Air Quality Supervision

Lesson 7

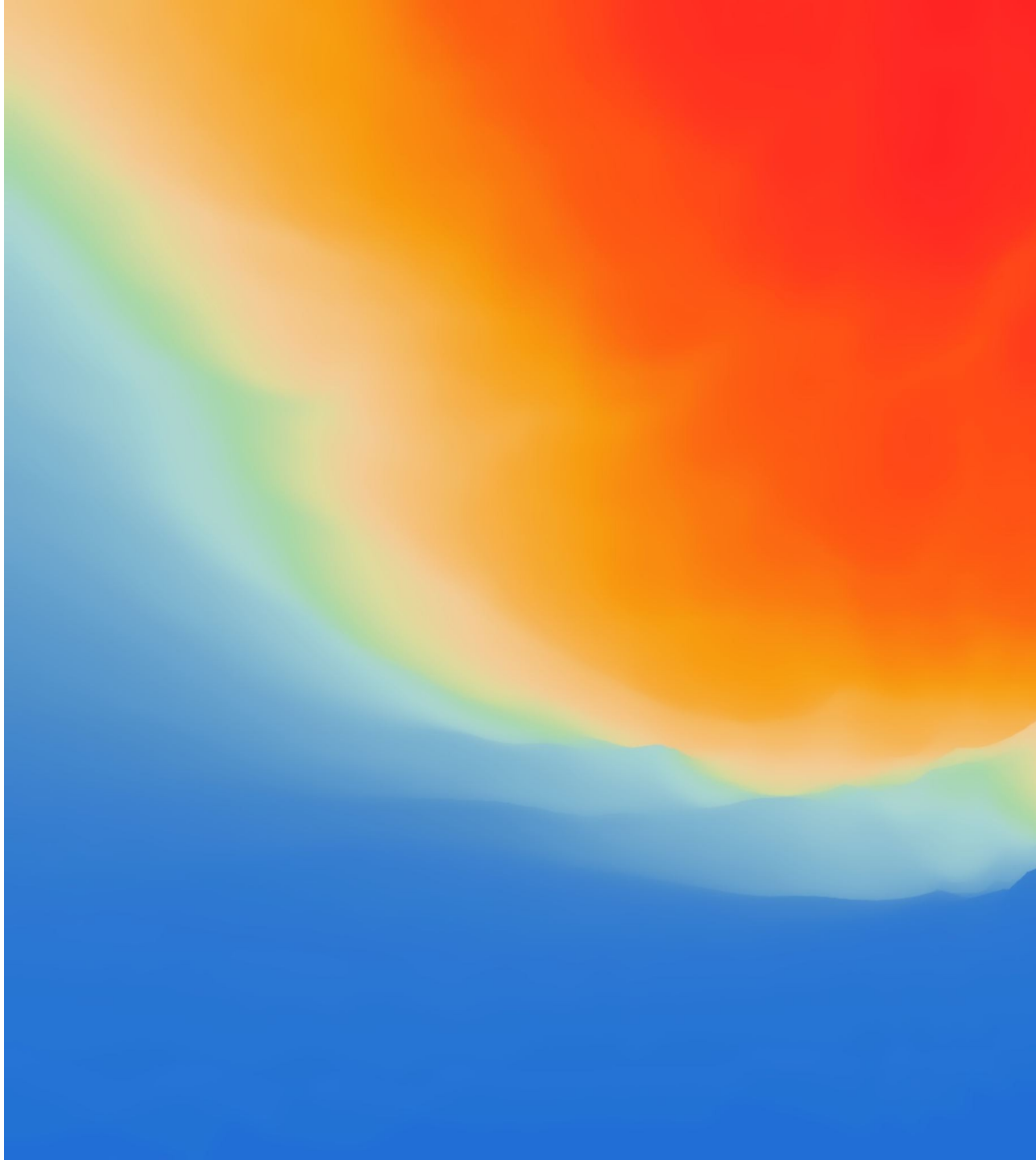
Objectives

At the end of the lesson, students should be able to:

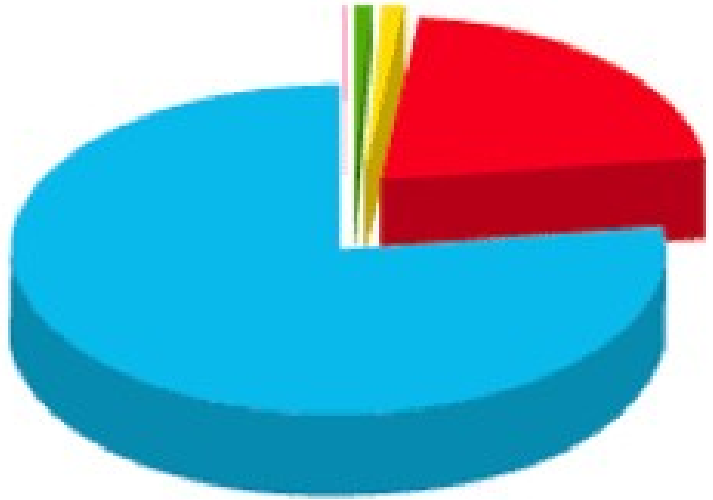
1. Define air and describe its composition.
2. Identify the major layers of the atmosphere.
3. Explain the importance of air to living organisms and the environment.
4. Describe common sources and effects of air pollution.
5. Explain major air-related environmental problems and their effects on humans and ecosystems.

Air

- **Air** is a **mixture of gases** that makes up the **Earth’s atmosphere**.



Atmospheric composition



- Nitrogen (N_2), 78.09%
- Oxygen (O_2), 20.95%
- Argon (Ar), 0.93%
- Carbon dioxide (CO_2), 0.038%
- Minute traces of neon (Ne), helium (He), methane (CH_4), water vapor (H_2O), krypton (Kr), hydrogen (H), xenon (Xe), and ozone (O_3).

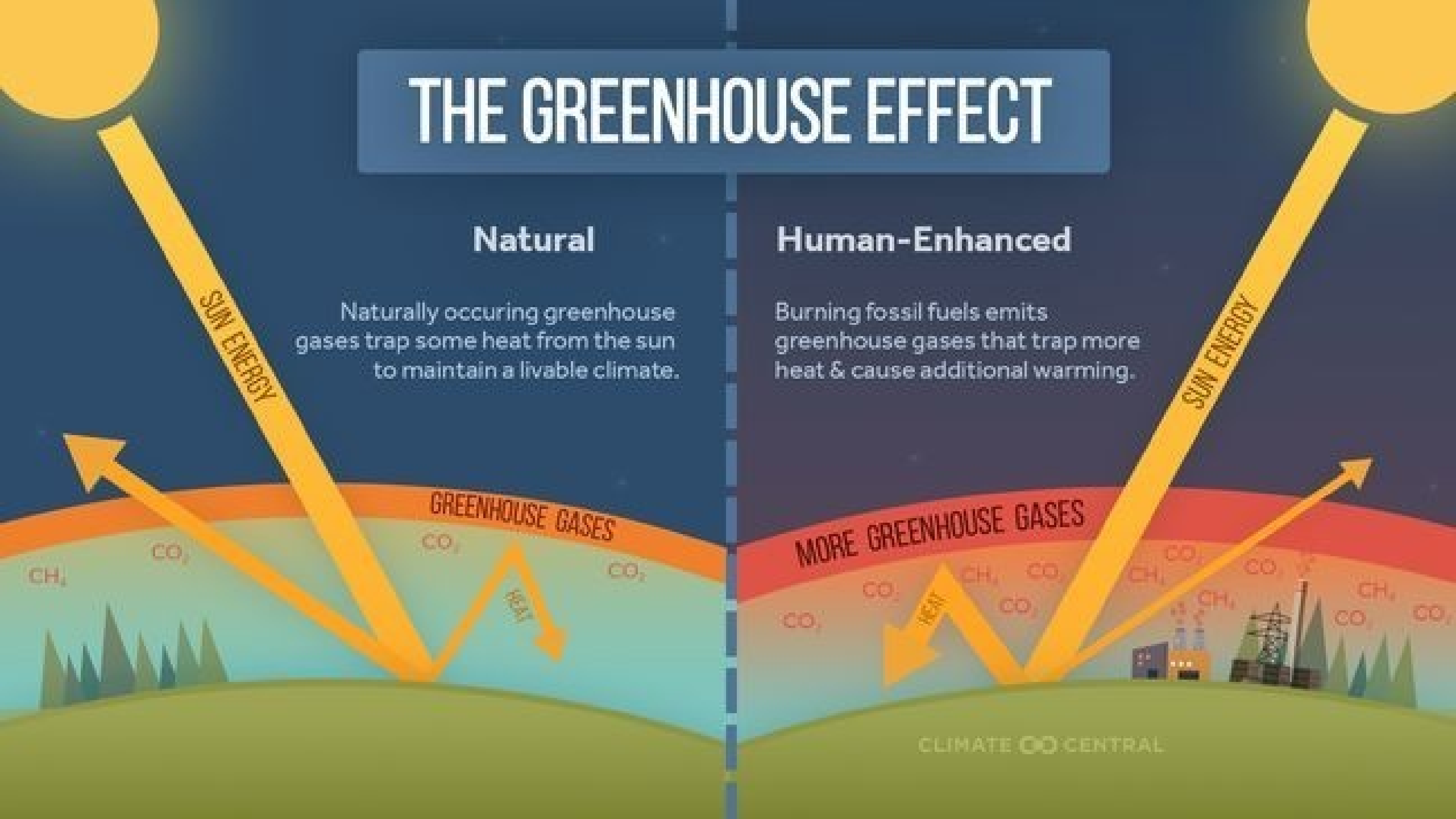
Composition of Air (Dry Air)

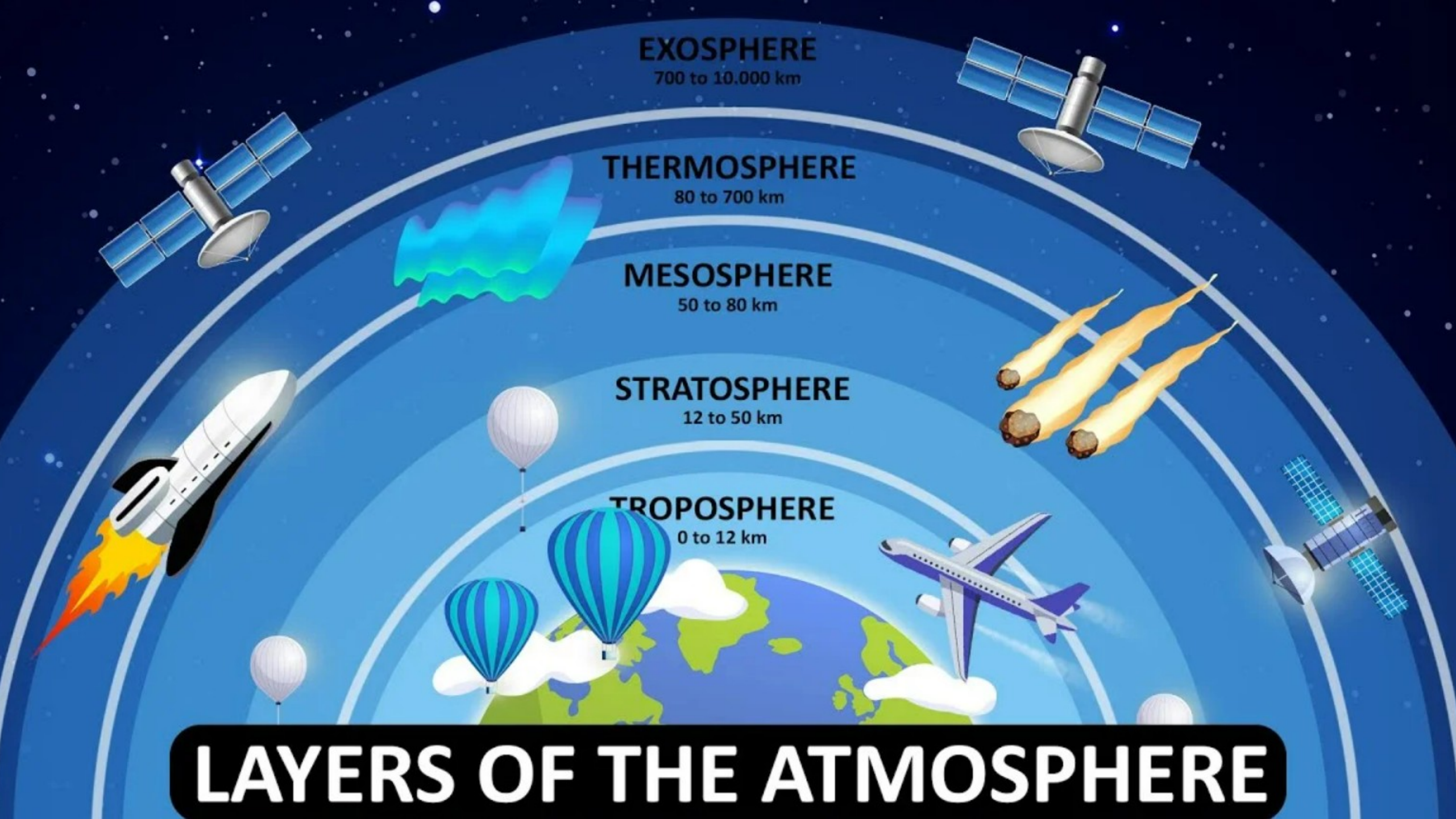
Gas	Percentage	
Nitrogen (N_2)	~78%	Needed for plant growth (nitrogen cycle). Dilute oxygen.
Oxygen (O_2)	~21%	Essential for respiration and combustion.
Argon (Ar)	~0.93%	Inert harmless gas.
Carbon Dioxide (CO_2)	~0.04%	Used in photosynthesis. Helps regulate Earth's temperature (greenhouse effect)
Other gases	Trace	Neon, helium, methane, krypton, hydrogen, xenon, **ozone,** etc. Plus, **water vapor** (varying by humidity) and tiny solid particles like dust, pollen, and smoke.

Greenhouse Effect

- The **greenhouse effect** is a **natural process** that **warms the Earth's surface**.
  - When the Sun's energy reaches the Earth's atmosphere, some of it is reflected to space, and the rest is absorbed and re-radiated by greenhouse gases.
  - These absorbed gases then radiate the heat back towards the Earth's surface, warming it.
  - This process is essential for life on Earth, as **it keeps the planet warm enough** to support living organisms. **Without the greenhouse effect, the Earth would be much colder.**
- **Human activities** are significantly **increasing the concentration of greenhouse gases**, leading to an **enhanced greenhouse effect** causing the planet to **warm too much.**

THE GREENHOUSE EFFECT




EXOSPHERE
700 to 10.000 km
THERMOSPHERE
80 to 700 km
MESOSPHERE
50 to 80 km
STRATOSPHERE
12 to 50 km
TROPOSPHERE
0 to 12 km
LAYERS OF THE ATMOSPHERE

Layers of the Atmosphere

Layer	Key Characteristics
Troposphere	Contains most of the air and water vapor. **Weather occurs here.**
Stratosphere	Contains the **ozone layer.** Protects life from harmful UV radiation.
Mesosphere	Coldest atmospheric layer **Meteors burn up here.**
Thermosphere	Very high temperatures **Auroras occur here.** Contains the ionosphere (important for radio communication)
Exosphere	Extremely thin air. **Where satellites orbit.**

Summary (Importance of Air)

- Supplies **oxygen for breathing (respiration).**
- Provides **carbon dioxide for photosynthesis.**
- Regulates **Earth’s temperature.**
  - The atmosphere acts like a blanket, keeping Earth warm (**greenhouse effect**)
- Enables **weather and climate.**
  - Air movements create wind, storms, and weather patterns.
- Protects Earth from **harmful solar radiation** (ozone layer).
  - Ozone layer filters harmful UV radiation
- Helps in **pollination and seed dispersal.**
- **Burns up most meteoroids** before they hit Earth's surface.



Air pollution

Topic 2

Air pollution

- **Air pollution** refers to the presence of **harmful or excessive quantities of substances in the air** that can have adverse effects on human health, the environment, and the climate.
- **Pollutants** are substances that contaminate the air and cause pollution.

Primary Pollutants

Primary Pollutants – Directly emitted into the atmosphere from sources.

1. **Carbon Monoxide (CO)** – Produced by the incomplete combustion of **fuels.**
2. **Sulfur Dioxide (SO_2)** – Emitted by burning coal and industrial processes.
3. **Nitrogen Oxides (NO_x)** – Released from **vehicles** and power plants.
4. Particulate Matter (PM) – From dust, smoke, and industrial emissions.
5. **Volatile Organic Compounds (VOCs)** – Found in **vehicle emissions** and solvents.

Secondary Pollutants

Secondary Pollutants – Formed when primary pollutants **react** in the atmosphere.

1. **Tropospheric Ozone (O_3)** - formed when NO_x and VOCs react with sunlight.
2. **Smog** - a mixture of pollutants, including ground-level ozone and particulates.
3. **Acid rain** - formed from SO_2 and NO_x mixing with water vapor in the atmosphere.

Sources of Air Pollution

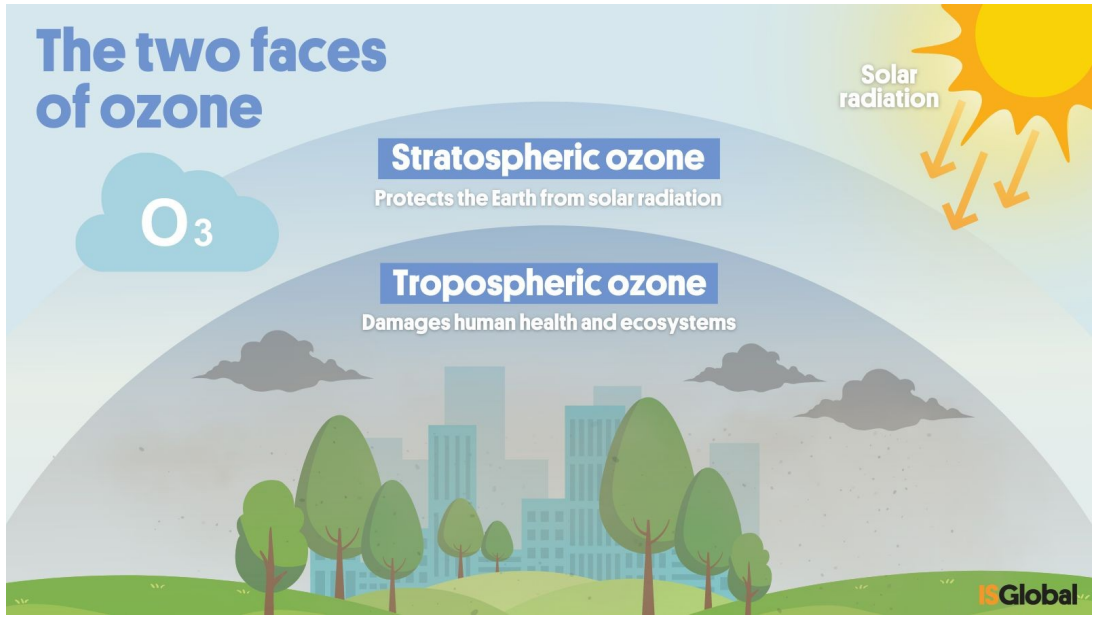
Natural Sources

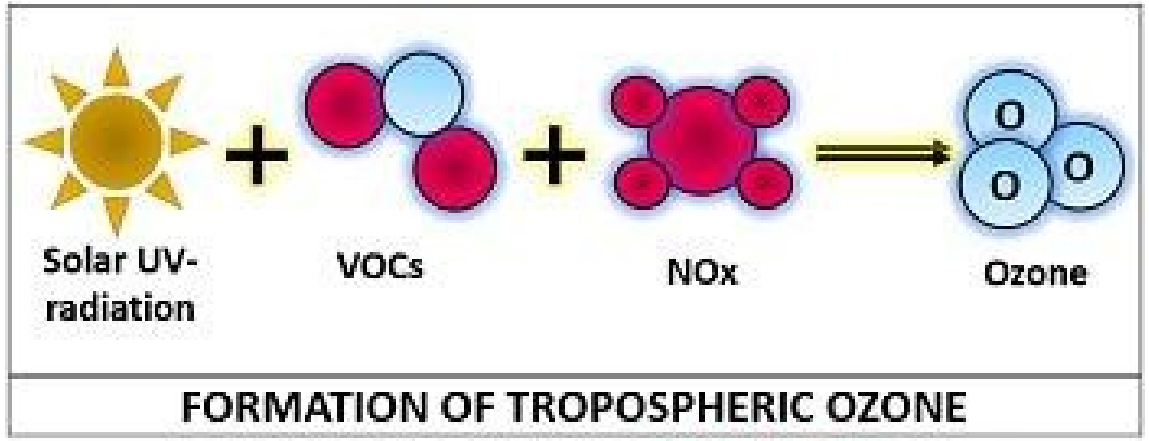
1. **Volcanic eruptions**
   - emit SO_2, ash, and particulate matter.
2. **Wildfires**
   - produce CO, NOx, and particulate matter.
3. **Dust storms**
   - release fine particulate matter into the air.
4. **Biological decay**
   - releases methane and other gases.

Sources of Air Pollution

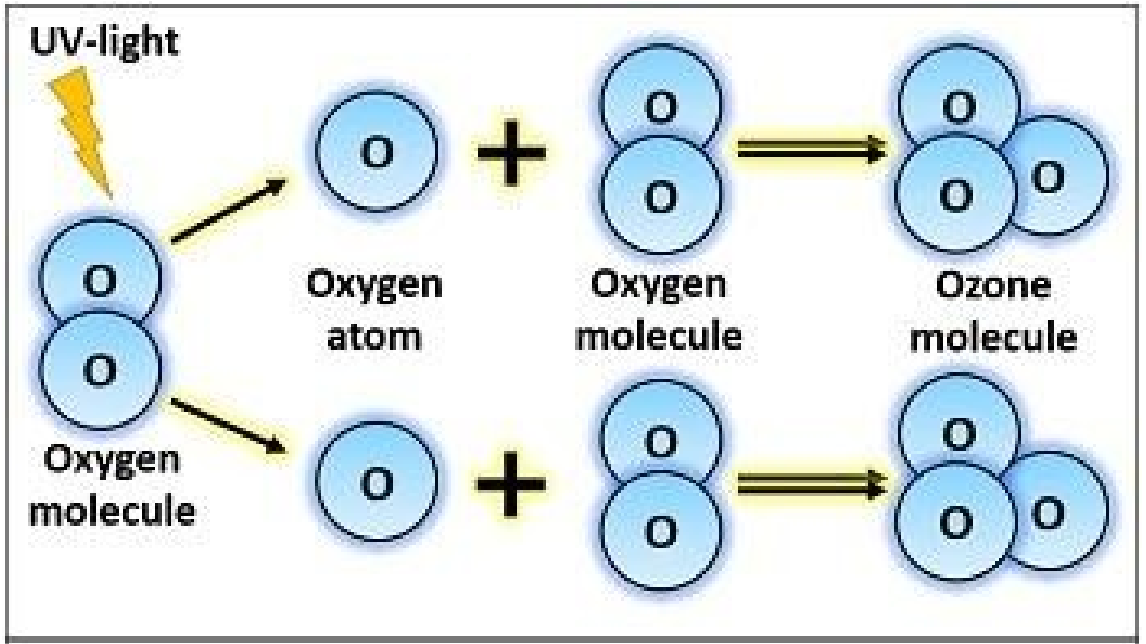
Human-Made Sources

1. **Industrial emissions**
   - factories release SO_2, NO_x, and heavy metals.
2. **Vehicle emissions**
   - cars and trucks emit carbon monoxide, nitrogen oxides, and hydrocarbons.
3. **Burning of fossil fuels**
   - coal, oil, and natural gas combustion releases CO_2, SO_2, and NO_x.
4. **Agricultural activities**
   - fertilizers and pesticides release ammonia and CH4.
5. **Deforestation**
   - reduces oxygen production and increases CO_2 levels.





FORMATION OF TROPOSPHERIC OZONE



FORMATION OF STRATOSPHERIC OZONE

Air-Related Environmental Problems

Topic 3

- **Global Warming**
- **Ozone Layer Depletion**
- **Acid Rain**

Global Warming

Global warming is the gradual increase in Earth's average temperature due to the buildup of **greenhouse gases** in the atmosphere.

Key Greenhouse Gases:

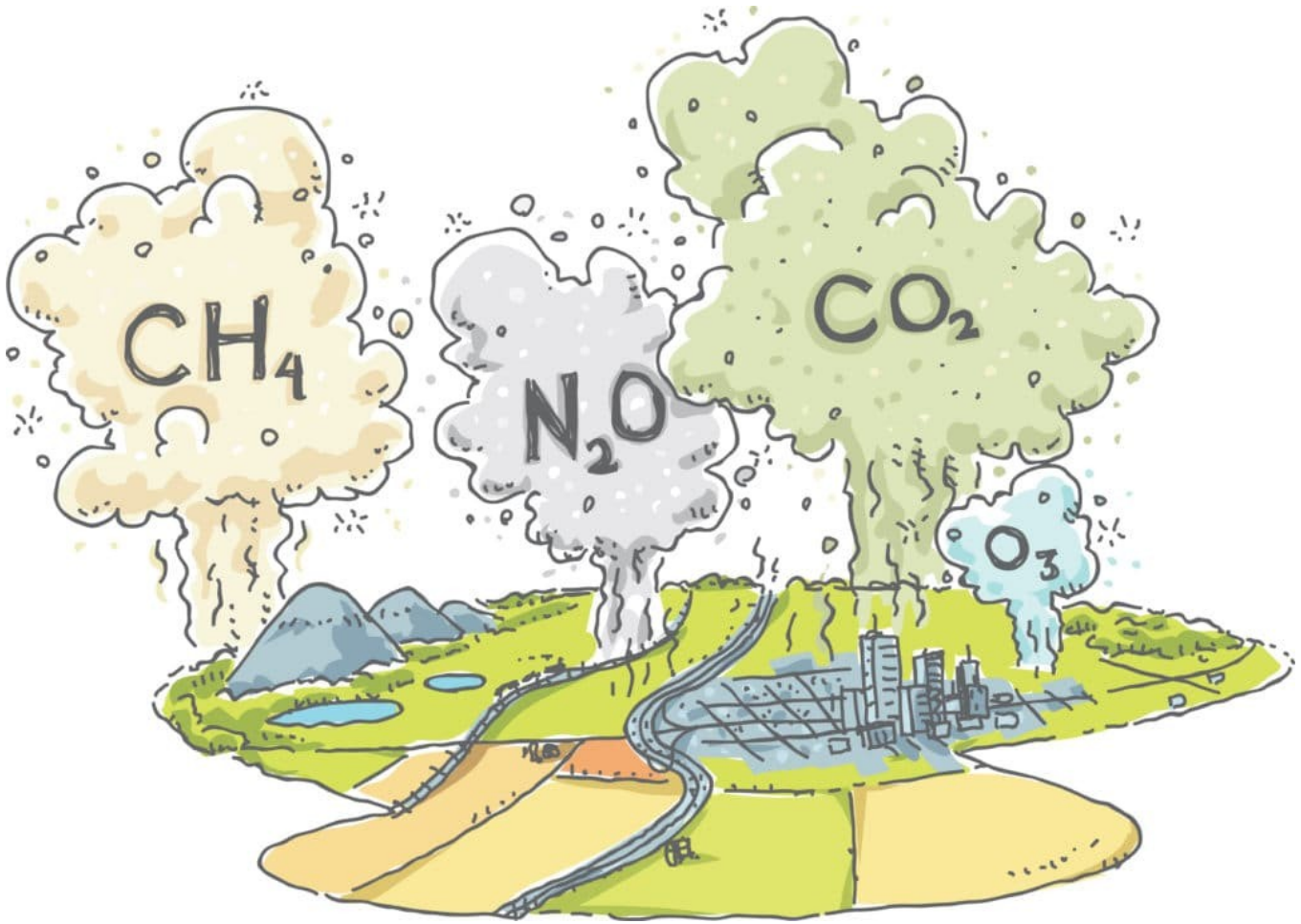
- Carbon dioxide (CO_2)
- Methane (CH_4)
- Nitrous oxide (N_2O)

Causes:

- Burning of fossil fuels
- Deforestation
- Industrial activities

Effects:

- Rising sea levels
- Stronger typhoons and heat waves
- Climate change affecting agriculture and ecosystems


CH_4
N_2O
CO_2
O_3





Ozone Layer Depletion

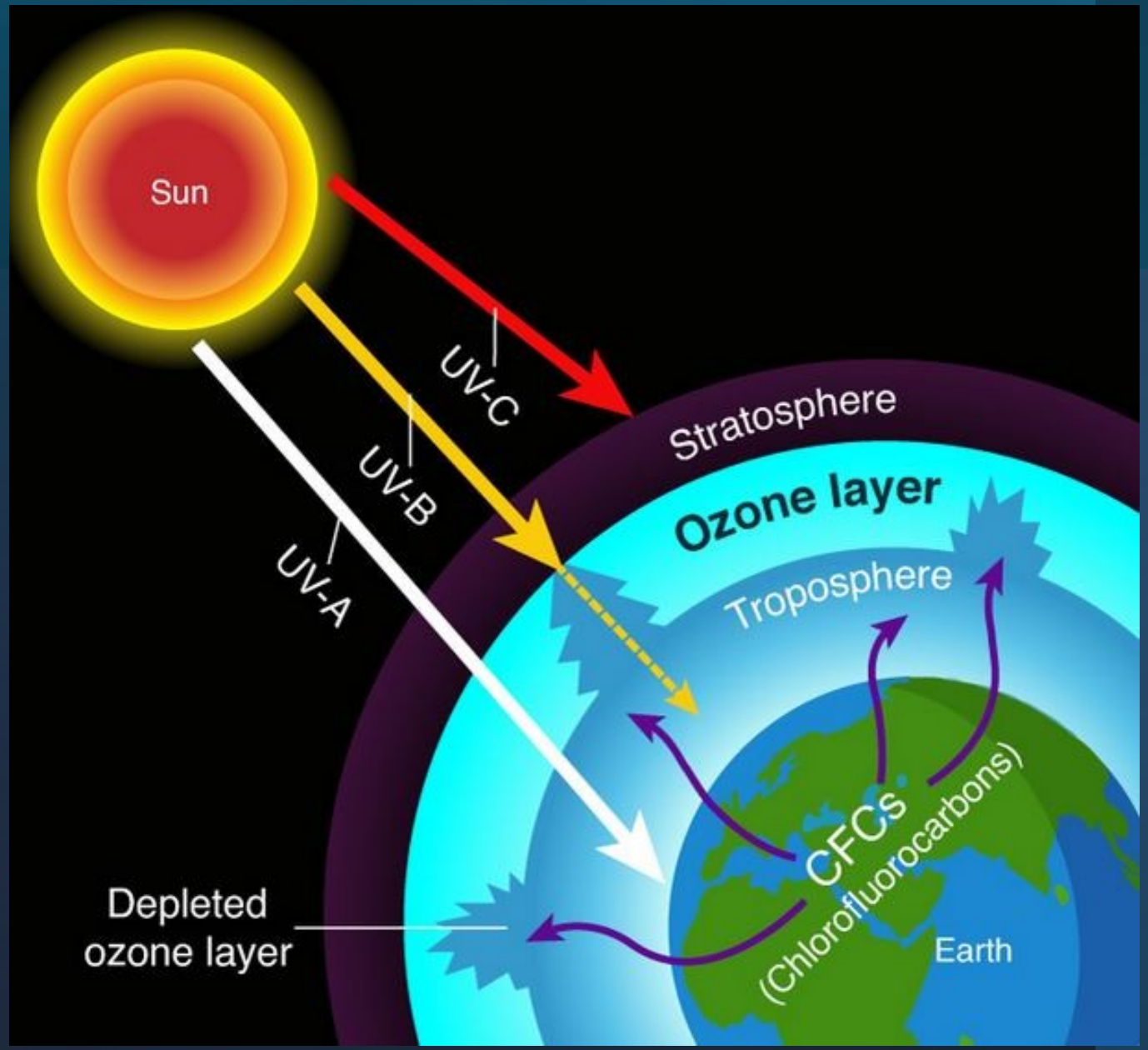
The **ozone layer** is a part of the stratosphere that protects Earth from harmful **ultraviolet (UV) radiation**.

Causes:

- Chlorofluorocarbons (CFCs)
- Industrial chemicals and aerosols

Effects:

- Increased skin cancer risk
- Damage to crops and marine life
- Eye problems in humans


Sun
UV-C
UV-B
UV-A
Stratosphere
Ozone layer
Troposphere
CFCs
(Chlorofluorocarbons)
Earth
Depleted
ozone layer

Acid Rain

Acid rain refers to rain, snow, or fog that contains strong acids formed from air pollutants.

Main Pollutants:

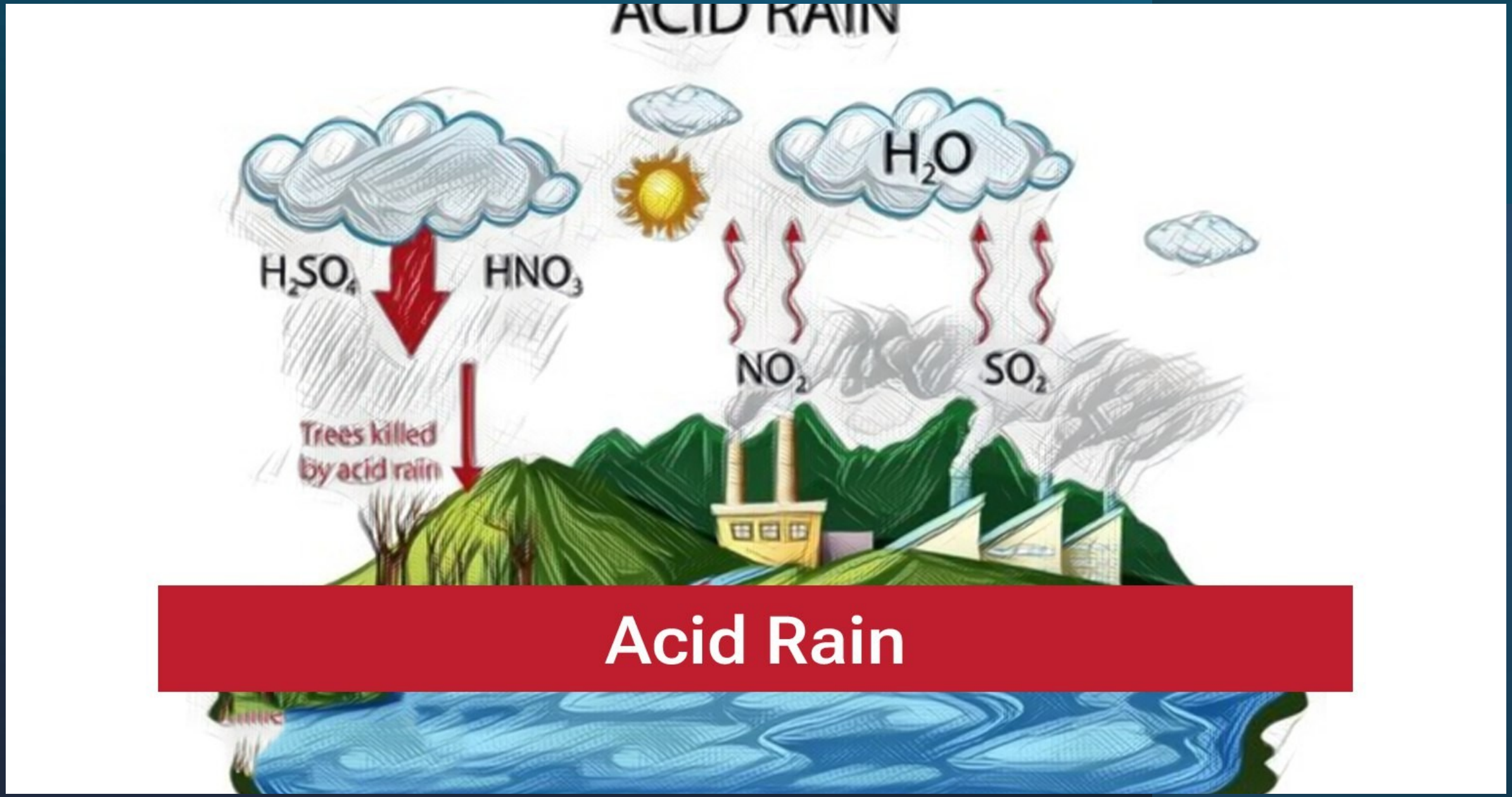
- Sulfur dioxide (SO_2)
- Nitrogen oxides (NO_x)

Sources:

- Power plants
- Vehicle emissions
- Factory smoke

Effects:

- Damage to forests
- Fish death in lakes and rivers
- Corrosion of buildings and monuments



Acid Rain

Air Pollution Control Methods

Topic 4

Air Pollution Control Methods are techniques and strategies used to **reduce, prevent, or remove pollutants from the air**.

A. Source Control

—

- Prevent pollution **before it enters the air**.

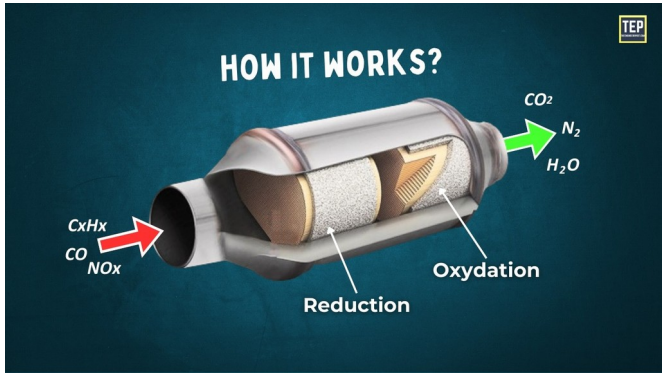
Examples:

- Using **clean and renewable energy** (solar, wind).
- Fuel-efficient and electric vehicles.
- Proper maintenance of machines and vehicles.
- Reducing fossil fuel use.

B. Engineering and Technological Controls

Devices that **remove pollutants** from emissions.

Device	Function
Electrostatic Precipitator	Removes dust and smoke particles from factory exhaust
Scrubber	Uses liquid to remove SO_2 and other gases
Catalytic Converter	Reduces harmful gases from vehicle exhaust
Filters	Trap particulate matter



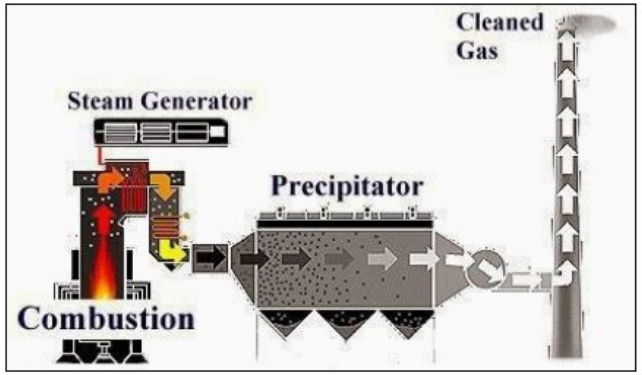
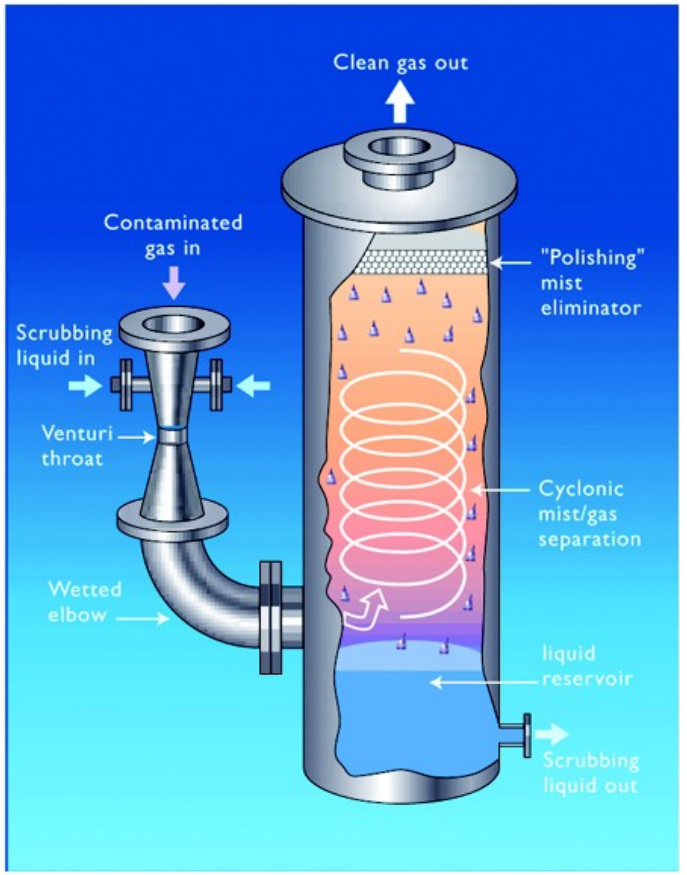
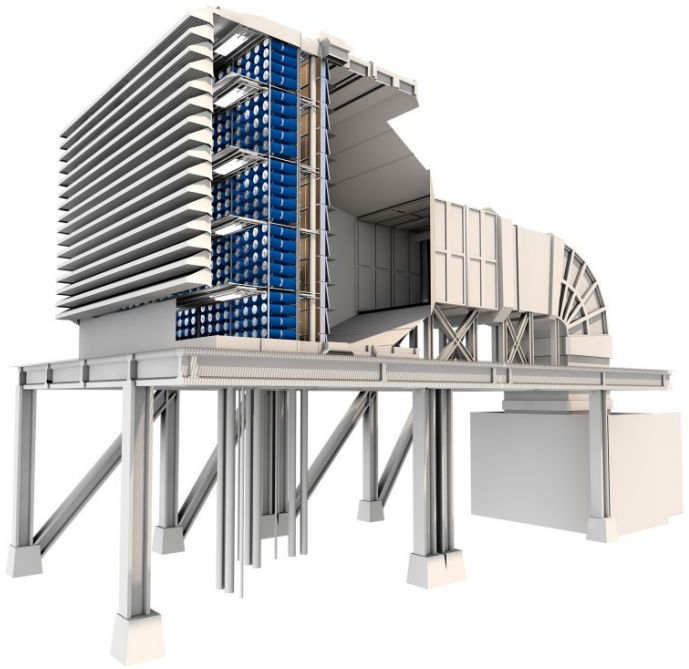


Figure 1: ESP application in a combustion power plant [3].





C. Legal and Policy Measure

- Government actions to control pollution.

Examples:

- Emission standards
- Anti-smoke belching programs
- Ban on open burning
- Environmental laws (e.g., **Philippine Clean Air Act – RA 8749**)

Philippine Clean Air Act – RA 8749

- The **Philippine Clean Air Act of 1999 (RA 8749)** is a law that aims to **protect and preserve the country's air quality** by controlling air pollution from various sources. Its main goal is to ensure **clean and safe air for the health and well-being of Filipinos**.

Regulations:

- Sets **ambient air quality standards** for key pollutants.
- Requires emission standards for vehicles and industries.
- Mandates the use of cleaner fuels and environmentally friendly technologies.

D. Individual and Community Actions

Everyone plays a role in keeping the air clean.

- Planting trees.
- Proper waste disposal.
- Using public transportation.
- Conserving electricity.
- Avoiding the burning of trash.